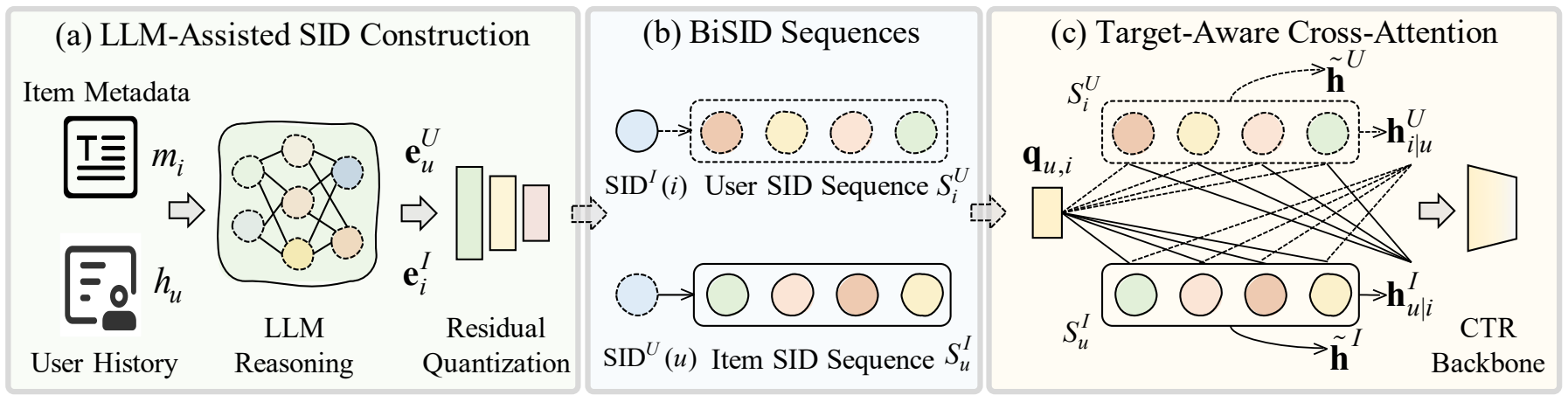

(a) LLM-Assisted SID Construction
Item Metadata
m_i
h_u
User History
LLM Reasoning
$\mathbf{e}_u^U$
$\mathbf{e}_i^I$
Residual Quantization
(b) BiSID Sequences
$\mathrm{SID}^I(i)$ User SID Sequence S_i^U
$\mathrm{SID}^U(u)$ Item SID Sequence S_u^I
(c) Target-Aware Cross-Attention
S_i^U
$\tilde{\mathbf{h}}^U$
$\mathbf{h}_{i|u}^U$
$\mathbf{q}_{u,i}$
S_u^I
$\mathbf{h}_{u|i}^I$
$\tilde{\mathbf{h}}^I$
CTR Backbone